

6

PHASE PLANE

6.0 Introduction

This chapter begins our study of two-dimensional *nonlinear* systems. First we consider some of their general properties. Then we classify the kinds of fixed points that can arise, building on our knowledge of linear systems (Chapter 5). The theory is further developed through a series of examples from biology (competition between two species) and physics (conservative systems, reversible systems, and the pendulum). The chapter concludes with a discussion of index theory, a topological method that provides global information about the phase portrait.

This chapter is mainly about fixed points. The next two chapters will discuss closed orbits and bifurcations in two-dimensional systems.

6.1 Phase Portraits

The general form of a vector field on the phase plane is

$$\begin{aligned}\dot{x}_1 &= f_1(x_1, x_2) \\ \dot{x}_2 &= f_2(x_1, x_2)\end{aligned}$$

where f_1 and f_2 are given functions. This system can be written more compactly in vector notation as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$$

where $\mathbf{x} = (x_1, x_2)$ and $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}))$. Here $\mathbf{x}$ represents a point in the phase plane, and $\dot{\mathbf{x}}$ is the velocity vector at that point. By flowing along the vector field,

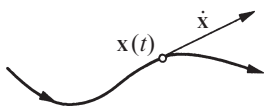


Figure 6.1.1

a phase point traces out a solution $\mathbf{x}(t)$, corresponding to a trajectory winding through the phase plane (Figure 6.1.1). Furthermore, the entire phase plane is filled with trajectories, since each point can play the role of an initial condition.

For nonlinear systems, there's typically no hope of finding the trajectories analytically. Even when explicit formulas are available, they are often too complicated to provide much insight. Instead we will try to determine the *qualitative* behavior of the solutions. Our goal is to find the system's phase portrait directly from the properties of $\mathbf{f}(\mathbf{x})$. An enormous variety of phase portraits is possible; one example is shown in Figure 6.1.2.

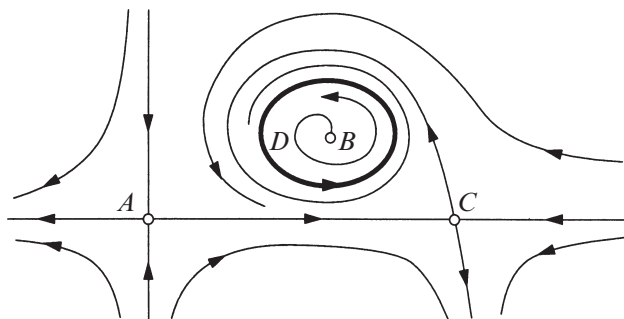


Figure 6.1.2

Some of the most salient features of any phase portrait are:

1. The **fixed points**, like A , B , and C in Figure 6.1.2. Fixed points satisfy $\mathbf{f}(\mathbf{x}^*) = \mathbf{0}$, and correspond to steady states or equilibria of the system.
2. The **closed orbits**, like D in Figure 6.1.2. These correspond to periodic solutions, i.e., solutions for which $\mathbf{x}(t+T) = \mathbf{x}(t)$ for all t , for some $T > 0$.
3. The arrangement of trajectories near the fixed points and closed orbits. For example, the flow pattern near A and C is similar, and different from that near B .
4. The stability or instability of the fixed points and closed orbits. Here, the fixed points A , B , and C are unstable, because nearby trajectories tend to move away from them, whereas the closed orbit D is stable.

Numerical Computation of Phase Portraits

Sometimes we are also interested in *quantitative* aspects of the phase portrait. Fortunately, numerical integration of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ is not much harder than that of $\dot{x} = f(x)$. The numerical methods of Section 2.8 still work, as long as we replace

the numbers x and $f(x)$ by the vectors $\mathbf{x}$ and $\mathbf{f}(\mathbf{x})$. We will always use the Runge-Kutta method, which in vector form is

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$$

where

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_n)\Delta t$$

$$\mathbf{k}_2 = \mathbf{f}(\mathbf{x}_n + \frac{1}{2}\mathbf{k}_1)\Delta t$$

$$\mathbf{k}_3 = \mathbf{f}(\mathbf{x}_n + \frac{1}{2}\mathbf{k}_2)\Delta t$$

$$\mathbf{k}_4 = \mathbf{f}(\mathbf{x}_n + \mathbf{k}_3)\Delta t.$$

A stepsize $\Delta t = 0.1$ usually provides sufficient accuracy for our purposes.

When plotting the phase portrait, it often helps to see a grid of representative vectors in the vector field. Unfortunately, the arrowheads and different lengths of the vectors tend to clutter such pictures. A plot of the **direction field** is clearer: short line segments are used to indicate the local direction of flow.

EXAMPLE 6.1.1:

Consider the system $\dot{x} = x + e^{-y}$, $\dot{y} = -y$. First use qualitative arguments to obtain information about the phase portrait. Then, using a computer, plot the direction field. Finally, use the Runge-Kutta method to compute several trajectories, and plot them on the phase plane.

Solution: First we find the fixed points by solving $\dot{x} = 0$, $\dot{y} = 0$ simultaneously. The only solution is $(x^*, y^*) = (-1, 0)$. To determine its stability, note that $y(t) \rightarrow 0$ as $t \rightarrow \infty$, since the solution to $\dot{y} = -y$ is $y(t) = y_0 e^{-t}$. Hence $e^{-y} \rightarrow 1$ and so in the long run, the equation for x becomes $\dot{x} \approx x + 1$; this has exponentially growing solutions, which suggests that the fixed point is unstable. In fact, if we restrict our attention to initial conditions on the x -axis, then $y_0 = 0$ and so $y(t) = 0$ for all time. Hence the flow on the x -axis is governed *strictly* by $\dot{x} = x + 1$. Therefore the fixed point is unstable.

To sketch the phase portrait, it is helpful to plot the **nullclines**, defined as the curves where either $\dot{x} = 0$ or $\dot{y} = 0$. The nullclines indicate where the flow is purely horizontal or vertical (Figure 6.1.3). For example, the flow is horizontal where $\dot{y} = 0$, and since $\dot{y} = -y$, this occurs on the line $y = 0$. Along this line, the flow is to the right where $\dot{x} = x + 1 > 0$, that is, where $x > -1$.

Similarly, the flow is vertical where $\dot{x} = x + e^{-y} = 0$, which occurs on the curve shown in Figure 6.1.3. On the upper part of the curve where $y > 0$, the flow is downward, since $\dot{y} < 0$.

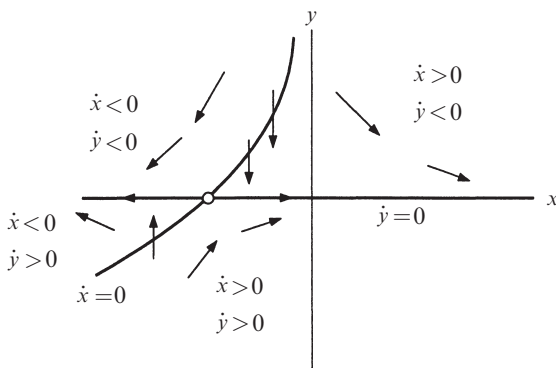


Figure 6.1.3

The nullclines also partition the plane into regions where $\dot{x}$ and $\dot{y}$ have various signs. Some of the typical vectors are sketched above in Figure 6.1.3. Even with the limited information obtained so far, Figure 6.1.3 gives a good sense of the overall flow pattern.

Now we use the computer to finish the problem. The direction field is indicated by the line segments in Figure 6.1.4, and several trajectories are shown. Note how the trajectories always follow the local slope.

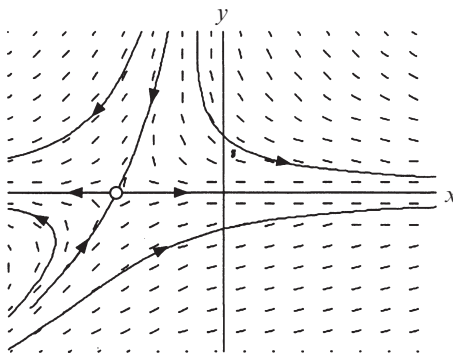


Figure 6.1.4

The fixed point is now seen to be a nonlinear version of a saddle point. ■

6.2 Existence, Uniqueness, and Topological Consequences

We have been a bit optimistic so far—at this stage, we have no guarantee that the general nonlinear system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ even *has* solutions! Fortunately the existence and uniqueness theorem given in Section 2.5 can be generalized to two-dimensional

systems. We state the result for n -dimensional systems, since no extra effort is involved:

Existence and Uniqueness Theorem: Consider the initial value problem $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, $\mathbf{x}(0) = \mathbf{x}_0$. Suppose that $\mathbf{f}$ is continuous and that all its partial derivatives $\partial f_i / \partial x_j$, $i, j = 1, \dots, n$, are continuous for $\mathbf{x}$ in some open connected set $D \subset \mathbf{R}^n$. Then for $\mathbf{x}_0 \in D$, the initial value problem has a solution $\mathbf{x}(t)$ on some time interval $(-\tau, \tau)$ about $t = 0$, and the solution is unique.

In other words, existence and uniqueness of solutions are guaranteed if $\mathbf{f}$ is continuously differentiable. The proof of the theorem is similar to that for the case $n = 1$, and can be found in most texts on differential equations. Stronger versions of the theorem are available, but this one suffices for most applications.

From now on, we'll assume that all our vector fields are smooth enough to ensure the existence and uniqueness of solutions, starting from any point in phase space.

The existence and uniqueness theorem has an important corollary: *different trajectories never intersect*. If two trajectories *did* intersect, then there would be two solutions starting from the same point (the crossing point), and this would violate the uniqueness part of the theorem. In more intuitive language, a trajectory can't move in two directions at once.

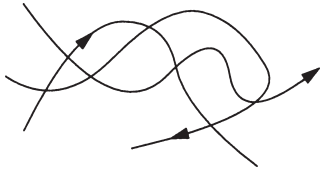


Figure 6.2.1

Because trajectories can't intersect, phase portraits always have a well-groomed look to them. Otherwise they might degenerate into a snarl of criss-crossed curves (Figure 6.2.1). The existence and uniqueness theorem prevents this from happening.

In two-dimensional phase spaces (as opposed to higher-dimensional phase spaces), these results have especially strong topological consequences. For example, suppose there is a closed orbit C in the phase plane. Then any trajectory starting inside C is trapped in there forever (Figure 6.2.2).

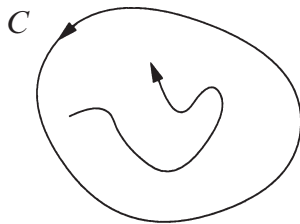


Figure 6.2.2

What is the fate of such a bounded trajectory? If there are fixed points inside C , then of course the trajectory might eventually approach one of them. But what if there *aren't* any fixed points? Your intuition may tell you that the trajectory can't meander around forever—if so, you're right. For vector fields on the plane, the **Poincaré-Bendixson theorem** states that if a trajectory is confined to a closed, bounded region and there are no fixed points in the region, then the trajectory must eventually approach a closed orbit. We'll discuss this important theorem in Section 7.3.

But that part of our story comes later. First we must become better acquainted with fixed points.

6.3 Fixed Points and Linearization

In this section we extend the *linearization* technique developed earlier for one-dimensional systems (Section 2.4). The hope is that we can approximate the phase portrait near a fixed point by that of a corresponding linear system.

Linearized System

Consider the system

$$\dot{x} = f(x, y)$$

$$\dot{y} = g(x, y)$$

and suppose that (x^*, y^*) is a fixed point, i.e.,

$$f(x^*, y^*) = 0, \quad g(x^*, y^*) = 0.$$

Let

$$u = x - x^*, \quad v = y - y^*$$

denote the components of a small disturbance from the fixed point. To see whether the disturbance grows or decays, we need to derive differential equations for u and v . Let's do the u -equation first:

$$\dot{u} = \dot{x} \quad (\text{since } x^* \text{ is a constant})$$

$$= f(x^* + u, y^* + v) \quad (\text{by substitution})$$

$$= f(x^*, y^*) + u \frac{\partial f}{\partial x} + v \frac{\partial f}{\partial y} + O(u^2, v^2, uv) \quad (\text{Taylor series expansion})$$

$$= u \frac{\partial f}{\partial x} + v \frac{\partial f}{\partial y} + O(u^2, v^2, uv) \quad (\text{since } f(x^*, y^*) = 0).$$

To simplify the notation, we have written $\partial f / \partial x$ and $\partial f / \partial y$, but please remember that these partial derivatives are to be evaluated *at the fixed point* (x^*, y^*) ; thus they are *numbers*, not functions. Also, the shorthand notation $O(u^2, v^2, uv)$ denotes **quadratic terms** in u and v . Since u and v are small, these quadratic terms are *extremely* small.

Similarly we find

$$\dot{v} = u \frac{\partial g}{\partial x} + v \frac{\partial g}{\partial y} + O(u^2, v^2, uv).$$

Hence the disturbance (u, v) evolves according to

$$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} + \text{quadratic terms.} \quad (1)$$

The matrix

$$A = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}_{(x^*, y^*)}$$

is called the **Jacobian matrix** at the fixed point (x^*, y^*) . It is the multivariable analog of the derivative $f'(x^*)$ seen in Section 2.4.

Now since the quadratic terms in (1) are tiny, it's tempting to neglect them altogether. If we do that, we obtain the **linearized system**

$$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} \quad (2)$$

whose dynamics can be analyzed by the methods of Section 5.2.

The Effect of Small Nonlinear Terms

Is it really safe to neglect the quadratic terms in (1)? In other words, does the linearized system give a qualitatively correct picture of the phase portrait near (x^*, y^*) ? The answer is *yes, as long as the fixed point for the linearized system is not one of the borderline cases* discussed in Section 5.2. In other words, if the linearized system predicts a saddle, node, or a spiral, then the fixed point *really is* a saddle, node, or spiral for the original nonlinear system. See Andronov et al. (1973) for a proof of this result, and Example 6.3.1 for a concrete illustration.

The borderline cases (centers, degenerate nodes, stars, or non-isolated fixed points) are much more delicate. They can be altered by small nonlinear terms, as we'll see in Example 6.3.2 and in Exercise 6.3.11.

EXAMPLE 6.3.1:

Find all the fixed points of the system $\dot{x} = -x + x^3$, $\dot{y} = -2y$, and use linearization to classify them. Then check your conclusions by deriving the phase portrait for the full nonlinear system.

Solution: Fixed points occur where $\dot{x} = 0$ and $\dot{y} = 0$ simultaneously. Hence we need $x = 0$ or $x = \pm 1$, and $y = 0$. Thus, there are three fixed points: $(0, 0)$, $(1, 0)$, and $(-1, 0)$. The Jacobian matrix at a general point (x, y) is

$$A = \begin{pmatrix} \frac{\partial \dot{x}}{\partial x} & \frac{\partial \dot{x}}{\partial y} \\ \frac{\partial \dot{y}}{\partial x} & \frac{\partial \dot{y}}{\partial y} \end{pmatrix} = \begin{pmatrix} -1 + 3x^2 & 0 \\ 0 & -2 \end{pmatrix}.$$

Next we evaluate A at the fixed points. At $(0, 0)$, we find $A = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}$, so $(0, 0)$ is a stable node. At $(\pm 1, 0)$, $A = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$, so both $(1, 0)$ and $(-1, 0)$ are saddle points.

Now because stable nodes and saddle points are not borderline cases, we can be certain that the fixed points for the full nonlinear system have been predicted correctly.

This conclusion can be checked explicitly for the nonlinear system, since the x and y equations are *uncoupled*; the system is essentially two independent first-order systems at right angles to each other. In the y -direction, all trajectories decay exponentially to $y = 0$. In the x -direction, the trajectories are attracted to $x = 0$ and repelled from $x = \pm 1$. The vertical lines $x = 0$ and $x = \pm 1$ are *invariant*, because $\dot{x} = 0$ on them; hence any trajectory that starts on these lines stays on them forever. Similarly, $y = 0$ is an invariant horizontal line. As a final observation, we note that the phase portrait must be symmetric in both the x and y axes, since the equations are invariant under the transformations $x \rightarrow -x$ and $y \rightarrow -y$. Putting all this information together, we arrive at the phase portrait shown in Figure 6.3.1.

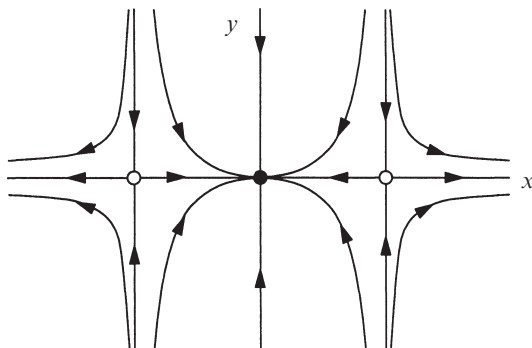


Figure 6.3.1

This picture confirms that $(0, 0)$ is a stable node, and $(\pm 1, 0)$ are saddles, as expected from the linearization. ■

The next example shows that small nonlinear terms can change a center into a spiral.

EXAMPLE 6.3.2:

Consider the system

$$\begin{aligned}\dot{x} &= -y + ax(x^2 + y^2) \\ \dot{y} &= x + ay(x^2 + y^2)\end{aligned}$$

where a is a parameter. Show that the linearized system *incorrectly* predicts that the origin is a center for all values of a , whereas in fact the origin is a stable spiral if $a < 0$ and an unstable spiral if $a > 0$.

Solution: To obtain the linearization about $(x^*, y^*) = (0, 0)$, we can either compute the Jacobian matrix directly from the definition, or we can take the following shortcut. For any system with a fixed point at the origin, x and y represent deviations from the fixed point, since $u = x - x^* = x$ and $v = y - y^* = y$; hence we can linearize by simply omitting nonlinear terms in x and y . Thus the linearized system is $\dot{x} = -y$, $\dot{y} = x$. The Jacobian is

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

which has $\tau = 0$, $\Delta = 1 > 0$, so the origin is always a center, according to the linearization.

To analyze the nonlinear system, we change variables to **polar coordinates**. Let $x = r \cos \theta$, $y = r \sin \theta$. To derive a differential equation for r , we note $x^2 + y^2 = r^2$, so $x\dot{x} + y\dot{y} = r\dot{r}$. Substituting for $\dot{x}$ and $\dot{y}$ yields

$$\begin{aligned}r\dot{r} &= x(-y + ax(x^2 + y^2)) + y(x + ay(x^2 + y^2)) \\ &= a(x^2 + y^2)^2 \\ &= ar^4.\end{aligned}$$

Hence $\dot{r} = ar^3$. In Exercise 6.3.12, you are asked to derive the following differential equation for θ :

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2}.$$

After substituting for $\dot{x}$ and $\dot{y}$ we find $\dot{\theta} = 1$. Thus in polar coordinates the original system becomes

$$\begin{aligned}\dot{r} &= ar^3 \\ \dot{\theta} &= 1.\end{aligned}$$

The system is easy to analyze in this form, because the radial and angular motions are independent. All trajectories rotate about the origin with constant angular velocity $\dot{\theta} = 1$.

The radial motion depends on a , as shown in Figure 6.3.2.

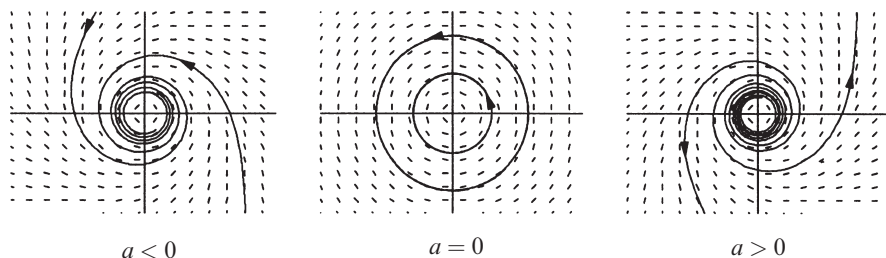


Figure 6.3.2

If $a < 0$, then $r(t) \rightarrow 0$ monotonically as $t \rightarrow \infty$. In this case, the origin is a stable spiral. (However, note that the decay is extremely slow, as suggested by the computer-generated trajectories shown in Figure 6.3.2.) If $a = 0$, then $r(t) = r_0$ for all t and the origin is a center. Finally, if $a > 0$, then $r(t) \rightarrow \infty$ monotonically and the origin is an unstable spiral.

We can see now why centers are so delicate: all trajectories are required to close *perfectly* after one cycle. The slightest miss converts the center into a spiral. ■

Similarly, stars and degenerate nodes can be altered by small nonlinearities, but unlike centers, *their stability doesn't change*. For example, a stable star may be changed into a stable spiral (Exercise 6.3.11) but not into an unstable spiral. This is plausible, given the classification of linear systems in Figure 5.2.8: stars and degenerate nodes live squarely in the stable or unstable region, whereas centers live on the razor's edge between stability and instability.

If we're only interested in *stability*, and not in the detailed geometry of the trajectories, then we can classify fixed points more coarsely as follows:

Robust cases:

Repellers (also called *sources*): both eigenvalues have positive real part.

Attractors (also called *sinks*): both eigenvalues have negative real part.

Saddles: one eigenvalue is positive and one is negative.

Marginal cases:

Centers: both eigenvalues are pure imaginary.

Higher-order and non-isolated fixed points: at least one eigenvalue is zero.

Thus, from the point of view of stability, the marginal cases are those where at least one eigenvalue satisfies $\text{Re}(\lambda) = 0$.

Hyperbolic Fixed Points, Topological Equivalence, and Structural Stability

If $\operatorname{Re}(\lambda) \neq 0$ for both eigenvalues, the fixed point is often called *hyperbolic*. (This is an unfortunate name—it sounds like it should mean “saddle point”—but it has become standard.) Hyperbolic fixed points are sturdy; their stability type is unaffected by small nonlinear terms. Nonhyperbolic fixed points are the fragile ones.

We’ve already seen a simple instance of hyperbolicity in the context of vector fields on the line. In Section 2.4 we saw that the stability of a fixed point was accurately predicted by the linearization, *as long as* $f'(x^*) \neq 0$. This condition is the exact analog of $\operatorname{Re}(\lambda) \neq 0$.

These ideas also generalize neatly to higher-order systems. A fixed point of an n th-order system is *hyperbolic* if all the eigenvalues of the linearization lie off the imaginary axis, i.e., $\operatorname{Re}(\lambda_i) \neq 0$ for $i = 1, \dots, n$. The important *Hartman-Grobman theorem* states that the local phase portrait near a hyperbolic fixed point is “topologically equivalent” to the phase portrait of the linearization; in particular, the stability type of the fixed point is faithfully captured by the linearization. Here *topologically equivalent* means that there is a *homeomorphism* (a continuous deformation with a continuous inverse) that maps one local phase portrait onto the other, such that trajectories map onto trajectories and the sense of time (the direction of the arrows) is preserved.

Intuitively, two phase portraits are topologically equivalent if one is a distorted version of the other. Bending and warping are allowed, but not ripping, so closed orbits must remain closed, trajectories connecting saddle points must not be broken, etc.

Hyperbolic fixed points also illustrate the important general notion of structural stability. A phase portrait is *structurally stable* if its topology cannot be changed by an arbitrarily small perturbation to the vector field. For instance, the phase portrait of a saddle point is structurally stable, but that of a center is not: an arbitrarily small amount of damping converts the center to a spiral.

6.4 Rabbits versus Sheep

In the next few sections we’ll consider some simple examples of phase plane analysis. We begin with the classic *Lotka-Volterra model of competition* between two species, here imagined to be rabbits and sheep. Suppose that both species are competing for the same food supply (grass) and the amount available is limited. Furthermore, ignore all other complications, like predators, seasonal effects, and other sources of food. Then there are two main effects we should consider:

1. Each species would grow to its carrying capacity in the absence of the other. This can be modeled by assuming logistic growth for each species (recall Section 2.3). Rabbits have a legendary ability to reproduce, so perhaps we should assign them a higher intrinsic growth rate.